

Chinese Remainder Theorem

Assignment-4:

$$a) x \equiv 1 \pmod{3}, x \equiv 2 \pmod{5}, x \equiv 3 \pmod{7}$$

The moduli 3, 5, 7 are coprime, so CRT applies.

$$\text{Let } m = 3 \times 5 \times 7 = 105$$

m_i	a_i	$m_i = m/m_i$	Inverse of $m_i \pmod{m_i}$
3	1	35	$35 \equiv 2 \pmod{3} \rightarrow 2$
5	2	21	$21 \equiv 1 \pmod{5} \rightarrow 1$
7	3	15	$15 \equiv 1 \pmod{7} \rightarrow 1$

$$x \equiv \sum a_i m_i y_i \pmod{m}$$

$$x \equiv 1(35)(2) + 2(21)(1) + 3(15)(1) \pmod{m}$$

$$\equiv 70 + 42 + 45 \pmod{m}$$

$$\equiv 157 \pmod{105}$$

$$\equiv 52$$

$$\therefore x \equiv 52 \pmod{105}$$

$$ii) x \equiv 5 \pmod{11}, x \equiv 14 \pmod{29}, x \equiv 15 \pmod{31}$$

(all moduli are coprime)

$$m = 11 \times 29 \times 31 = 9889$$

m_i	a_i	$m_i' = \frac{m}{m_i}$	inverse of $m_i' \pmod{m_i}$
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11	5	899	$899 \equiv 8 \pmod{11} \rightarrow 7$
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29	14	341	$341 \equiv 22 \pmod{29} \rightarrow 9$
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31	15	319	$319 \equiv 9 \pmod{31} \rightarrow 7$
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$$x \equiv 5(899)(7) + 14(341)(9) + 15(319)(7)$$

$$= 31465 + 43096 + 33495$$

$$\equiv 84056 \pmod{9889}$$

$$= 4944$$

$$\therefore x = 4944 \pmod{9889}$$

iii) $x \equiv 5 \pmod{6}, x \equiv 4 \pmod{11}, x \equiv 3 \pmod{17}$

$$M = 6 \times 11 \times 17 = 1122$$

m_i	a_i	$M_i = M/m_i$	inverse of $M_i \pmod{m_i}$
6	5	187	$187 \equiv 1 \pmod{6} \Rightarrow 1$
11	4	102	$103 \equiv 3 \pmod{11} \rightarrow 4$
17	3	66	$66 \equiv 15 \pmod{17} \rightarrow 8$

$$x \equiv 5(187)(1) + 4(102)(4) + 3(66)(8)$$

$$= 935 + 1632 + 1584$$

$$\equiv 4151 \pmod{1122}$$

$$= 785$$

$$\therefore x \equiv 785 \pmod{1122}$$